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# Patent Certificate

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SAARTHI GREENTECH PRIVATE LIMITED

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It is hereby certified that a patent has been granted to the patentee for an invention entitled ***AN APPARATUS FOR HYDROGEN GENERATION AND A METHOD THEREOF*** as disclosed in the above mentioned application for the term of 20 years from the 26<sup>th</sup> day of February 2024 in accordance with the provisions of the Patents Act, 1970.

It is hereby certified that a patent has been granted to the patentee for an invention entitled ***AN APPARATUS FOR HYDROGEN GENERATION AND A METHOD THEREOF*** as disclosed in the above mentioned application for the term of 20 years from the 26<sup>th</sup> day of February 2024 in accordance with the provisions of the Patents Act, 1970.



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**Note.** - The fees for renewal of this patent, if it is to be maintained, will fall / has fallen due on 26<sup>th</sup> day of February 2026 and on the same day in every year thereafter.

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**FORM 2**

THE PATENTS ACT, 1970 (39 of 1970) & THE PATENTS RULES, 2003

COMPLETE SPECIFICATION

(See section 10 and rule 13)

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**AN APPARATUS FOR HYDROGEN GENERATION AND A METHOD  
THEREOF**

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The following specification particularly describes the disclosure and the manner  
in which is to be performed.



## 5 FIELD OF INVENTION

[0001] Embodiments of the present disclosure relates to gas generation devices and more particularly to an apparatus for hydrogen generation and a method thereof.

## BACKGROUND

[0002] Many useful gases are produced by electrolysis using an electrolysis gas  
10 generating device including an electrolysis chamber. Hydrogen and oxygen are common examples of useful gases that can be generated by electrolyzing water in the form of a conductive aqueous solution. The awareness of the usefulness of hydrogen is increasing and the hydrogen is considered as a clean energy source. The electrolytic gas generating devices may have received attention as a cost-  
15 effective and important devices for generating hydrogen.

[0003] Although it is known that the generation of useful gases such as hydrogen and oxygen by electrolysis is very cost-effective, in the process of generation of the hydrogen and oxygen gas, explosion may occur if the hydrogen is mixed with oxygen. The above problem limits the direct application of the electrolysis in gas  
20 generating devices in many applications such as automobiles, due to safety concerns. Presently, there is lack of on-demand generation and delivery of significant quantities of Hydrogen gas to internal combustion engine. This leads to generation of additional carbon footprint. There is need of a device that considerably reduces fuel utilization and detrimental emissions.

25 [0004] Hence, there is a need for an apparatus for hydrogen generation and a method thereof which addresses the aforementioned issues.

## OBJECTIVE OF THE INVENTION

[0005] An objective of the present invention is to provide an apparatus for hydrogen gas generation in a cost-effective way.

5 [0006] Another objective of the present invention is to facilitate energy harnessing from the engine associated battery and alternator with no additional carbon footprint.

[0007] Yet, an objective of the present invention is to achieve a considerable reductions in fuel utilization and detrimental emissions in a vehicle.

10 [0008] Further, an objective of the present invention is to generate hydrogen gas and delivers of significant quantities of Hydrogen gas to internal combustion (IC) engine.

#### BRIEF DESCRIPTION

[0009] In accordance with an embodiment of the present disclosure, an apparatus  
15 for hydrogen generation is provided. The apparatus includes a hydrogen generating unit and a venting pipe. The hydrogen generating unit includes an electrolyte tank. The electrolyte tank includes an electrolyte supply line, an electrolyser assembly, a turbulence inducing unit, a reaction chamber, a manifold, a plastic chamber, and a power supply unit. The electrolyte supply line and an electrolyte return line is  
20 positioned at a base of the electrolyte tank. The electrolyte supply line is adapted to transfer electrolyte to the electrolyte water tank. The electrolyser assembly connected to the electrolyte supply line and adapted to pass the electrolyte through a plurality of electrode plates to the electrolyte water tank in a turbulent manner via an exit nipple. The plurality of electrode plates cover a pre-defined area with a pre-  
25 defined spacing between two consecutive electrode plates. Each electrode plate of the plurality of electrode plates comprises a predefined surface area. Each of the plurality of electrode plates comprises an aperture at a corner of the electrode plate to reduce the charge concentration at the corners of the electrode plates and a central aperture. The central aperture is supported by a plurality of symmetrical apertures  
30 to bind the plurality of electrode plates in an array with a plurality of bolts. The plurality of electrode plates is assembled in a plurality of configurations of a plurality of cells, wherein a predetermined cells of the plurality of cells are adapted to connect with an anode and a cathode. The turbulence inducing unit connected

5 with the electrolyser assembly. The turbulence inducing unit is a combination of a high-pressure water flow arrangement and a flow reducing diverter plate. The flow reducing diverter plate comprises a plurality of first predetermined holes on two sides and a plurality of second predetermined holes beneath a middle of the flow reducing diverter plate. The reaction chamber includes a moulded polypropylene plastic is connected to the flow reducing diverter plate. The reaction chamber includes a plurality of electrolyte pressure chambers positioned beneath the flow reducing diverter plate. The manifold connected to the electrolyte supply line. The manifold is adapted to pump water at a predetermined pressure to create turbulence between the plurality of electrode plates. The plastic chamber connected with the manifold. The plastic chamber comprises a plurality of electrodes positioned on at least one of the same side or on opposite side of the electrolyser assembly. The plurality of the electrode plates are configured using double, single and double cells use same side electrodes and the electrode port provided on the opposite side of the plastic chamber is provided for a plurality of double cells configuration. The power supply unit connected with the electrolyte tank and configured to use a forced convection system and a predefined capacity water tank. The power supply is a direct current power supply with an anode point and a cathode point, arranged to pass a current of a pre-defined amperes to the system at a predefined current range. The hydrogen generating unit after a predefined period of constant operation, stabilizes at a pre-determined temperature with a predetermined amount of electrolyte consumption for each four hours. The venting pipe connected with the hydrogen generating unit, wherein the venting pipe is positioned atop of the hydrogen generating unit via an integrated electrolyte tank and a bubbler tank. The venting pipe is adapted to vent any gas escaping from the apparatus to the bubbler tank wherein the gas is scrubbed and vented on the exterior of the apparatus.

[0010] In accordance with another embodiment of the present disclosure, a method for operating the device for generating hydrogen is provided. The method includes transferring, by an electrolyte supply line of a hydrogen generating unit, electrolyte to an electrolyte water tank. The method also includes covering, a pre-defined area

5 with a pre-defined spacing between two consecutive electrode plates by the plurality of electrode plates, wherein each electrode plate of the plurality of electrode plates comprises a predefined surface area. Further, the method includes providing, a corner aperture and a central aperture at of each of the electrode plate of plurality of electrode plate, wherein the central aperture is supported by a  
10 plurality of symmetrical apertures to bind the plurality of electrode plates in an array with a plurality of bolts. Furthermore, the method includes assembling, the plurality of electrode plates in a plurality of configurations of a plurality of cells, wherein a predetermined cells of the plurality of cells are adapted to connect with an anode and a cathode. Furthermore, the method includes providing, the turbulence inducing  
15 unit as a combination of a high-pressure water flow arrangement and a flow reducing diverter plate. Furthermore, the method includes providing, a plurality of first predetermined holes on two sides and a plurality of second predetermined holes beneath a middle of the flow reducing diverter plate. Moreover, the method includes providing, a reaction chamber comprised of a moulded polypropylene plastic is  
20 connected to the flow reducing diverter plate, wherein the reaction chamber comprises a plurality of electrolyte pressure chambers positioned beneath the flow reducing diverter plate. Moreover, the method includes pumping, by a manifold, water at a predetermined pressure to create turbulence between the plurality of electrode plates. Moreover, the method includes positioning, a plurality of  
25 electrodes of a plastic chamber on at least one of the same side or on opposite side of the electrolyser assembly, wherein the plurality of the electrode plates using double, single, and double cells use same side electrodes and the electrode port provided on the opposite side of the plastic chamber. Moreover, the method includes supplying, by a power supply unit, power to use a forced convection  
30 system and a predefined capacity water tank, wherein the power supply is a direct current power supply with an anode point and a cathode point, wherein the anode point, and the cathode point is arranged to pass a current of a pre-defined amperes to the system at a predefined current range. Moreover, the method includes positioning, a venting pipe atop of the hydrogen generating unit via an integrated  
35 electrolyte tank and a bubbler tank, wherein the venting pipe is adapted to vent any

5 gas escaping from the apparatus to the bubbler tank wherein the gas is scrubbed and vented on the exterior of the apparatus.

[0011] To further clarify the advantages and features of the present disclosure, a more particular description of the disclosure will follow by reference to specific embodiments thereof, which are illustrated in the appended figures. It is to be  
10 appreciated that these figures depict only typical embodiments of the disclosure and are therefore not to be considered limiting in scope. The disclosure will be described and explained with additional specificity and detail with the appended figures.

#### BRIEF DESCRIPTION OF THE DRAWINGS

The disclosure will be described and explained with additional specificity and detail  
15 with the accompanying figures in which:

[0012] FIG. 1 is a schematic representation of an apparatus for hydrogen generation in accordance with an embodiment of the present disclosure;

[0013] FIG. 2a is a schematic representation of a side view of the apparatus for hydrogen generation of FIG. 1 in accordance with an embodiment of the present  
20 disclosure;

[0014] FIG. 2b is a schematic representation of a cross sectional front view of the apparatus for hydrogen generation of FIG. 1 in accordance with an embodiment of the present disclosure;

[0015] FIG. 2c is a schematic representation of a front view of the apparatus for  
25 hydrogen generation of FIG. 1 in accordance with an embodiment of the present disclosure;

[0016] FIG. 2d is a schematic representation of a top view of an electrode plate of FIG. 1 in accordance with an embodiment of the present disclosure;

- 5    [0017]    FIG. 3a is a schematic representation of an exemplary embodiment of the apparatus of hydrogen generation of FIG. 1 in accordance with an embodiment of the present disclosure;
- [0018]    FIG. 3b is a schematic representation a reaction chamber of FIG. 1 in accordance with an embodiment of the present disclosure;
- 10   [0019]    FIG. 3c is a schematic representation a top view of a plastic chamber of FIG. 1 in accordance with an embodiment of the present disclosure;
- [0020]    FIG. 3d is a schematic representation a cross sectional top view of a plastic chamber illustrating a reaction chamber of FIG. 1 in accordance with an embodiment of the present disclosure;
- 15   [0021]    FIG. 4a is a schematic representation a side view of an electrolyte water tank of FIG. 1 in accordance with an embodiment of the present disclosure;
- [0022]    FIG. 4b is a schematic representation a cross sectional view of a plurality of electrode plates of FIG. 1 in accordance with an embodiment of the present disclosure;
- 20   [0023]    FIG. 4c is a schematic representation of a flow reducing diverter plate of FIG. 1 in accordance with an embodiment of the present disclosure;
- [0024]    FIG. 5a is a flow chart representing the steps involved in a method for assembling the apparatus for hydrogen generation in accordance with an embodiment of the present disclosure; and
- 25   [0025]    FIG. 5b represents the continued steps involved in a method for assembling the apparatus for hydrogen generation of FIG. 5a in accordance with an embodiment of the present disclosure.
- [0026]    Further, those skilled in the art will appreciate that elements in the figures are illustrated for simplicity and may not have necessarily been drawn to scale.
- 30   Furthermore, in terms of the construction of the device, one or more components of



5 the device may have been represented in the figures by conventional symbols, and the figures may show only those specific details that are pertinent to understanding the embodiments of the present disclosure so as not to obscure the figures with details that will be readily apparent to those skilled in the art having the benefit of the description herein.

## 10 DETAILED DESCRIPTION

[0027] For the purpose of promoting an understanding of the principles of the disclosure, reference will now be made to the embodiment illustrated in the figures and specific language will be used to describe them. It will nevertheless be understood that no limitation of the scope of the disclosure is thereby intended.

15 Such alterations and further modifications in the illustrated system, and such further applications of the principles of the disclosure as would normally occur to those skilled in the art are to be construed as being within the scope of the present disclosure.

[0028] The terms "comprises", "comprising", or any other variations thereof, are

20 intended to cover a non-exclusive inclusion, such that a process or method that comprises a list of steps does not include only those steps but may include other steps not expressly listed or inherent to such a process or method. Similarly, one or more devices or sub-systems or elements or structures or components preceded by "comprises... a" does not, without more constraints, preclude the existence of other

25 devices, sub-systems, elements, structures, components, additional devices, additional sub-systems, additional elements, additional structures or additional components. Appearances of the phrase "in an embodiment", "in another embodiment" and similar language throughout this specification may, but not necessarily do, all refer to the same embodiment.

30 [0029] Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by those skilled in the art to which this disclosure belongs. The system, methods, and examples provided herein are only illustrative and not intended to be limiting.

5 [0030] In the following specification and the claims, reference will be made to a number of terms, which shall be defined to have the following meanings. The singular forms “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise.

[0031] Embodiments of the present disclosure relate to an apparatus for  
10 hydrogen generation. The apparatus includes a hydrogen generating unit and a venting pipe. The hydrogen generating unit includes an electrolyte tank. The electrolyte tank includes an electrolyte supply line, an electrolyser assembly, a turbulence inducing unit, a reaction chamber, a manifold, a plastic chamber, and a power supply unit. The electrolyte supply line and an electrolyte return line is  
15 positioned at a base of the electrolyte tank. The electrolyte supply line is adapted to transfer electrolyte to the electrolyte water tank. The electrolyser assembly is connected to the electrolyte supply line and adapted to pass the electrolyte through a plurality of electrode plates to the electrolyte water tank in a turbulent manner via an exit nipple. The plurality of electrode plates cover a pre-defined area with a pre-  
20 defined spacing between two consecutive electrode plates. Each electrode plate of the plurality of electrode plates comprises a predefined surface area. Each of the plurality of electrode plates comprises an aperture at a corner of the electrode plate to reduce the charge concentration at the corners of the electrode plates and a central aperture. The central aperture is supported by a plurality of symmetrical apertures  
25 to bind the plurality of electrode plates in an array with a plurality of bolts. The plurality of electrode plates is assembled in a plurality of configurations of a plurality of cells, wherein a predetermined cells of the plurality of cells are adapted to connect with an anode and a cathode. The turbulence inducing unit connected with the electrolyser assembly. The turbulence inducing unit is a combination of a  
30 high-pressure water flow arrangement and a flow reducing diverter plate. The flow reducing diverter plate comprises a plurality of first predetermined holes on two sides and a plurality of second predetermined holes beneath a middle of the flow reducing diverter plate. The reaction chamber includes a moulded polypropylene plastic is connected to the flow reducing diverter plate. The reaction chamber

5 includes a plurality of electrolyte pressure chambers positioned beneath the flow reducing diverter plate. The manifold connected to the electrolyte supply line. The manifold is adapted to pump water at a predetermined pressure to create turbulence between the plurality of electrode plates. The plastic chamber connected with the manifold. The plastic chamber comprises a plurality of electrodes positioned on at  
10 least one of the same side or on opposite side of the electrolyser assembly. The plurality of the electrode plates are configured using double, single and double cells use same side electrodes and the electrode port provided on the opposite side of the plastic chamber is provided for a plurality of double cells configuration. The power supply unit connected with the electrolyte tank and configured to use a forced  
15 convection system and a predefined capacity water tank. The power supply is a direct current power supply with an anode point and a cathode point, arranged to pass a current of a pre-defined amperes to the system at a predefined current range. The hydrogen generating unit, after a predefined period of constant operation, stabilizes at a pre-determined temperature with a predetermined amount of  
20 electrolyte consumption for each four hours. The venting pipe connected with the hydrogen generating unit, wherein the venting pipe is positioned atop of the hydrogen generating unit via an integrated electrolyte tank and a bubbler tank. The venting pipe is adapted to vent any gas escaping from the apparatus to the bubbler tank wherein the gas is scrubbed and vented on the exterior of the apparatus.

25 [0032] FIG. 1 is a schematic representation of an apparatus (100) for hydrogen generation in accordance with an embodiment of the present disclosure. The apparatus includes a hydrogen generating unit (102) and a venting pipe (134).

[0033] The hydrogen generating unit (102) includes an electrolyte water tank (104), an electrolyser assembly (110), a turbulence inducing unit (118, not shown  
30 in FIG. 1), a reaction chamber (124, shown in FIG. 3b), a manifold (126), a plastic chamber (128, shown in FIG. 3a), and a power supply unit (132). In one embodiment, the hydrogen generation unit (102) includes an upper diverter plate (142, shown in FIG. 3d) positioned at a top of the reaction chamber (124, shown in

5 FIG. 3b). The upper diverter plate (142, shown in FIG. 3d) includes a plurality of first apertures (116a, 116b, shown in FIG. 3d) larger than the plurality of holes.

[0034] The electrolyte water tank (104) includes an electrolyte supply line (106) and an electrolyser assembly (110). The electrolyte supply line (106) and an electrolyte return line (108) are positioned at a base of the electrolyte water tank  
10 (104). The electrolyte supply line (106) is adapted to transfer the electrolytes to the electrolyte water tank (104). In one embodiment, the electrolyte supply line (106) includes a filter adapted to be replaced or cleaned periodically.

[0035] The electrolyser assembly (110) is connected to the electrolyte supply line (106) and is adapted to pass the electrolyte through a plurality of electrode  
15 plates (112) to the electrolyte water tank (104) in a turbulent manner via an exit nipple (114). The plurality of electrode plates (112) cover a pre-defined area with a pre-defined spacing between two consecutive electrode plates, wherein each electrode plate of the plurality of electrode plates (112) includes a predefined surface area. Each of the plurality of electrode plates (112) includes an aperture  
20 (114a, shown in FIG. 2d) at a corner of the electrode plate to reduce the charge concentration at the corners of the electrode plates and a central aperture (114b, shown in FIG. 2d). The central aperture (114b, shown in FIG. 2d) is supported by a plurality of symmetrical apertures (114c, shown in FIG. 2d) to bind the plurality of electrode plates (112) in an array with a plurality of bolts (154, shown in FIG. 3c).  
25 The plurality of electrode plates (112) is assembled in a plurality of configurations of a plurality of cells. A predetermined cells of the plurality of cells are adapted to connect with an anode (120a) and a cathode (120b). In one embodiment, the plurality of electrode plates (112) is rectangular in shape and comprises a partial circular cutout (138a, shown in FIG. 2d) at three corners of the electrode plate (112)  
30 and one cutout (138b, shown in FIG.2d) at one remaining corner of the plate. In another embodiment, each of the electrode plate has an area of 18458 cm<sup>2</sup> with a spacing of 2 mm, for 577cm<sup>2</sup> surface area of each electrode plate.

5 [0036] The turbulence inducing unit is connected with the electrolyser assembly (120). The turbulence inducing unit is a combination of a high-pressure water flow arrangement and a flow reducing diverter plate (122, shown in FIG. 3d). Tthe flow reducing diverter plate (122, shown in FIG. 3d) includes a plurality of first predetermined holes on two sides and a plurality of second predetermined holes

10 (140b, shown in FIG. 4c) beneath a middle of the flow reducing diverter plate (122, shown in FIG. 3d). In one embodiment, the plurality of holes include a pipe with an inner diameter of 9 mm at top side of the pipe and one or more pipes of 4 mm inner diameter at the bottom side of the one or pipes for the flow of pressurized water through the reaction chamber (124, shown in FIG. 3b).

15 [0037] The reaction chamber (124) is typically fabricated from a moulded polypropylene plastic and is connected to the flow reducing diverter plate (122, shown in FIG. 3d). The reaction chamber (124) includes a plurality of electrolyte pressure chambers positioned beneath the flow reducing diverter plate (122).

[0038] The manifold (126) is connected to the electrolyte supply line (106). The

20 manifold (126) is adapted to pump water at a predetermined pressure to create turbulence between the plurality of electrode plates (112). In one embodiment, the manifold (126) is configured to clean and flush the plurality of electrolyte pressure chambers during service to clean a plurality of contaminants, wherein the plurality of contaminants comprises ferrous hydroxide and chromium hydroxide.

25 [0039] The plastic chamber (128, shown in FIG. 3a) is connected with the manifold (126). The plastic chamber (128) includes a plurality of electrodes (130, shown in FIG. 3b) is positioned on at least one of the same side or on opposite side of the electrolyser assembly (110). The plurality of the electrode plates (112) are configured using first double cell (130a, shown in FIG. 3b), single cell (130b,

30 shown in FIG. 3b) and double cells (130c, shown in FIG. 3b) use same side electrodes and the electrode port provided on the opposite side of the plastic chamber (128) is provided for a plurality of double cells configuration.

5 [0040] The power supply unit (132) is connected with the electrolyte water tank (104) and configured to use a forced convection system and a predefined capacity water tank. The power supply unit (132) is a direct current power supply with an anode point and a cathode point, arranged to pass a current of a pre-defined amperes to the system at a predefined current range. The hydrogen generating unit (102)  
10 after a predefined period of constant operation, stabilizes at a pre-determined temperature with a predetermined amount of electrolyte consumption for each four hours.

[0041] The venting pipe (134) is connected with the hydrogen generating unit (102). The venting pipe (134) is positioned atop of the hydrogen generating unit  
15 (102) via an integrated electrolyte tank and a bubbler tank (136). The venting pipe (134) is adapted to vent any gas escaping from the apparatus to the bubbler tank (136) wherein the gas is scrubbed and vented on the exterior of the apparatus.

[0042] In one embodiment, the apparatus (100) includes a de-humidifier connected with the bubbler tank (136) and adapted to remove moisture generated  
20 through the bubbler tank (136) to make the engine dry before the hydrogen gas reaches the engine. In one embodiment, the de-humidifier plays a crucial role in optimizing the performance and safety of the apparatus (100). As hydrogen output (HHO) gas is generated within the electrolyser assembly (110), it inherently carries moisture, especially since the water within the system can reach temperatures of 50  
25 to 80 degrees Celsius. This moisture-laden gas is then cycled back into the electrolyte water tank (104), where a separation process occurs. The water and gas are parted, with the water being retained in the tank and the HHO gas proceeding to the bubbler. The bubbler acts as a cooling and purifying stage, condensing steam back into liquid form, which inadvertently warms the water within the bubbler tank  
30 (136), leading to the generation of moist gas. The de-humidifier at this stage effectively removes moisture from the HHO gas, ensuring that the gas reaching the engine is dry. This step is vital not only for maintaining the integrity of the combustion process but also for preventing the corrosion and degradation of engine components that could occur due to excessive moisture. Through the de-humidifier,



5 the efficiency and longevity of the hydrogen generation unit is significantly enhanced, contributing to the overall efficacy of the HHO application in diesel engines.

[0043] FIG. 2a is a schematic representation of a side view of the apparatus for hydrogen generation of FIG. 1 in accordance with an embodiment of the present disclosure, FIG. 2b is a schematic representation of a cross sectional front view of the apparatus for hydrogen generation of FIG. 1 in accordance with an embodiment of the present disclosure and FIG. 2c is a schematic representation of a front view of the apparatus for hydrogen generation of FIG. 1 in accordance with an embodiment of the present disclosure. In one embodiment, potassium hydroxide neutralises the acidity of the hydrogen phosphate. In one embodiment, the apparatus (100) may be etched using an electrolyte comprising 3 litres of water mixed with 11 grams of potassium hydroxide for a period of 30 minutes at 15 to 25 amps then adding 25mls of phosphoric acid at a concentration of 185g/litre. In one embodiment, the apparatus (100) includes a water level indicator for bubbler (144) and a water level indication for electrolyte (146). In another embodiment, the apparatus allows to refill water. For refilling water, the apparatus (100) includes an electrolyte refill cap (148a) and a water refill cap (148b) for the bubbler tank (136).

[0044] In one embodiment, the water may be replenished through a plurality of refill caps while electrolyte need not be replenished every time unless the apparatus (100) gets cleaned and flushed. In one embodiment, the hydrogen generating unit (102) includes an electrolyte pump assembly located on an external wall of the generating unit. The electrolyte pump assembly is arranged to receive electrolyte from the electrolyte supply line (106) and to pump it to the manifold (126). The manifold (126) feeds into a base of the electrolyser assembly (110). In one embodiment, the venting pipe (134) is provided at the top of the hydrogen generating unit (102) via its integrated electrolyte and bubbler tank (136), ensuring that any gases which escape the apparatus are vented to the bubbler tank (136) where the gas is scrubbed and vented exterior of the hydrogen generating unit (102). As deemed, the gas can be introduced in air intake of exhaust pipes of vehicles or

5 in similar applications to aid in better fuel burning with hydrogen resulting in reduced fuel consumption and reduction of emission.

[0045] FIG. 2d is a schematic representation of a top view of an electrode plate of FIG. 1 in accordance with an embodiment of the present disclosure. In one embodiment, the plurality of electrode plates (112) may be rectangular in shape, with an aperture (114a) located in one corner and a central aperture (114b) supported by 4 symmetrical aperture (114c) to provide a perfect gap of desired dimension. The plurality of bolts (154, shown in FIG. 3c) ensures almost zero warpage of plates even in long usage. In addition, the electrode plates include cut outs (138a, 138b) located at appropriate intervals about their edges. This geometry allows the ready mounting of electrode plates (112) in a number of configurations, such as into a single cell (130b) or double cell (130c) configuration, simply by appropriate arrangement of mounting bolts. The partial circular cut outs (138a) and one corner cutout (138b) are made to the electrode plate, as the plate does not need clearance at all corners. This arrangement is done in order to reduce the charge concentration at the corners of the plates.

[0046] FIG. 3a is a schematic representation of an exemplary embodiment of the apparatus of hydrogen generation of FIG. 1 in accordance with an embodiment of the present disclosure. FIG. 3b is a schematic representation of a reaction chamber of FIG. 1 in accordance with an embodiment of the present disclosure. FIG. 3c is a schematic representation a top view of a plastic chamber of FIG. 1 in accordance with an embodiment of the present disclosure.

[0047] In one embodiment, the reaction chamber (124) is located within an open box which is preferably formed from insulating and chemical erosion resistant and heat bearing materials like polypropene plastic which is specifically moulded. In another embodiment, the reaction chamber (124) and an upper lid (156) are sealed with silicone O-ring and the tightened with M6 bolts. However, the present disclosure is not limited to the M6 bolts. An appropriate set of nut and bolts may be used based on a user requirement and design need. Yet in one embodiment, the

5 reaction chamber consists of three electrolysis enclosures arranged side-by-side. The electrolysis enclosures are separated by a plurality of separating plates. The separating plates are sized to locate within the reaction chamber (124) in a vertical orientation, and to substantially prevent fluid flow between adjacent enclosures. In one embodiment, the manifold (126) is attached to the exit nipple (114) into the  
10 pressure chambers. From here the electrolyte is urged upwards through the holes into the reaction chamber (124). The urging of the electrolyte causes it to travel upwards at a sufficient velocity to maintain turbulence along the entire face of the electrode plate assembly.

[0048] In one embodiment, the plastic chamber (128) consists of 2 electrodes  
15 on the same side or on opposite side of the electrolyser assembly which is plugged to the power supply unit (132) if not used. In one embodiment, the configuration of the electrode plate includes use of double cell (130a), single cell (130b) and double cells (130c). The double, single and double cells use same side electrodes while the opposite side is used by an electrode port of the plastic chamber (128) which is  
20 provided for 3 double cells configuration. In one embodiment, the cell has 11 plates with 3 positively charged plates, 3 negatively charged plates and 5 Neutral plates. The plastic chamber (128) includes an electrode hole which may be always plugged to the power unit.

[0049] FIG. 3d is a schematic representation a cross sectional top view of a  
25 reaction chamber of FIG. 1 in accordance with an embodiment of the present disclosure. In one embodiment the hydrogen generating unit (102) includes an upper diverter plate is located at the top of the reaction chamber (124). The upper diverter plate (142) includes a plurality of first apertures (116a, 116b) significantly larger than the holes through which electrolyte can flow substantially unimpeded.  
30 The hydrogen gas and oxygen gas is obtained by electrolysis of the electrolyte may also bubble through the holes of the upper diverter plate and can be collected in a gas collection area located above the reaction chamber (124). In one embodiment, the electrolyte passing through the upper diverter plate is allowed to flow out of the generating unit via the exit nipple (114) to the electrolyte water tank (104). In one

5     embodiment, the electrolyte may return through the electrolyte plate to the reaction chamber (124), but the apertures of the upper diverter plate prevents the 'backflow' of hydrogen gas and oxygen gas bubbles.

[0050]     FIG. 4a is a schematic representation a side view of an electrolyte water tank of FIG. 1 in accordance with an embodiment of the present disclosure. FIG. 10     4b is a schematic representation a cross sectional view of a plurality of electrode plates of FIG. 1 in accordance with an embodiment of the present disclosure. FIG. 4c is a schematic representation of a flow reducing diverter plate of FIG. 1 in accordance with an embodiment of the present disclosure. In one embodiment, the number electrode plate in the arrangement is 32 within the confined space of the 15     electrolysis assembly (110). The electrolysis assembly (110) forces the electrolyte to pass to the plurality of electrode plates (112), preferably in a turbulent manner. In one embodiment, the plurality of electrode plates (112), are typically made from titanium or stainless steel, and are in the order of 1.2mm thick.

[0051]     In a preferred embodiment, the plurality of electrode plates (112) are 20     made from high grade stainless steel with a relatively high molybdenum concentration been added. The potassium hydroxide acts to lower the surface tension of the electrolyte, which in turn reduces the size of gas bubbles generated and lowers the reactance of the system. Buffing the plates to a high mirror finish is optional to help detach the bubbles from the surface plates faster and lower the 25     reactance. The plurality of electrode plates (112) are spaced to produce a capacitive reactance in a freshwater electrolyte. In one embodiment, the space between the electrode plates (112) is provided by a partition plate (150). In another embodiment, each electrode plate has a spacing of 2 mm for 2100 cm<sup>2</sup> plates to achieve a desired capacitive resistance. This arrangement has plate area of 18458 cm<sup>2</sup> with a spacing 30     of 2 mm, for 577cm<sup>2</sup> surface area of each of the 32 plates. This gives a massive surface area for higher hydrogen production. In one embodiment, the electrode plates (112) may have nearly 1 mm to 3 mm spacing between each other may be used in the electrolyser assembly (110). In one embodiment, the flow reducing

5     diverter plate (122) includes a side notch (not shown in FIG. 4a, FIG.4b and FIG.4c)  
configured to fit in the turbulence inducing unit.

[0052]     FIG. 5a is a flow chart representing the steps involved in a method for  
assembling the apparatus for hydrogen generation in accordance with an  
embodiment of the present disclosure and FIG. 5b represents the continued steps  
10     involved in a method for assembling the apparatus for hydrogen generation of FIG.  
5a in accordance with an embodiment of the present disclosure.

[0053]     The method (200) includes transferring, by an electrolyte supply line of a  
hydrogen generating unit, electrolyte to an electrolyte water tank in step (202).

[0054]     The method (200) also includes covering, a pre-defined area with a pre-  
15     defined spacing between two consecutive electrode plates by the plurality of  
electrode plates, wherein each electrode plate of the plurality of electrode plates  
comprises a predefined surface area step (204). The method (200) also includes  
providing, a rectangular shaped plurality of electrode plates and including a partial  
circular cutout at three corners of the electrode plate and one cutout at one  
20     remaining corner of the plate.

[0055]     Further, the method (200) includes providing, a corner aperture and a central  
aperture at of each of the electrode plate of plurality of electrode plate, wherein the  
central aperture is supported by a plurality of symmetrical apertures to bind the  
plurality of electrode plates in an array with a plurality of bolts step (206).

25     [0056]     Furthermore, the method (200) includes assembling, the plurality of  
electrode plates in a plurality of configurations of a plurality of cells, wherein a  
predetermined cells of the plurality of cells are adapted to connect with an anode  
and a cathode step (208).

[0057]     Furthermore, the method (200) includes providing, the turbulence inducing  
30     unit as a combination of a high-pressure water flow arrangement and a flow  
reducing diverter plate step (210).

5 [0058] Furthermore, the method (200) includes providing, a plurality of first predetermined holes on two sides and a plurality of second predetermined holes beneath a middle of the flow reducing diverter plate step (212).

[0059] Moreover, the method (200) includes providing, a reaction chamber comprised of a moulded polypropylene plastic is connected to the flow reducing  
10 diverter plate, wherein the reaction chamber comprises a plurality of electrolyte pressure chambers positioned beneath the flow reducing diverter plate step (214).

[0060] Moreover, the method (200) includes pumping, by a manifold, water at a predetermined pressure to create turbulence between the plurality of electrode plates step (216). The method (200) also includes cleaning and flushing, the  
15 plurality of electrolyte pressure chambers during service to clean a plurality of contaminants, wherein the plurality of contaminants comprises ferrous hydroxide and chromium hydroxide.

[0061] Moreover, the method (200) includes positioning, a plurality of electrodes of a plastic chamber on at least one of the same side or on opposite side of the  
20 electrolyser assembly, wherein the plurality of the electrode plates using double, single and double cells use same side electrodes and the electrode port provided on the opposite side of the plastic chamber step (218).

[0062] Moreover, the method (200) includes supplying, by a power supply unit, power to use a forced convection system and a predefined capacity water tank,  
25 wherein the power supply is a direct current power supply with an anode point and a cathode point, wherein the anode point, and the cathode point is arranged to pass a current of a pre-defined amperes to the system at a predefined current range step (220).

[0063] Moreover, the method (200) includes positioning, a venting pipe atop of the  
30 hydrogen generating unit via an integrated electrolyte tank and a bubbler tank, wherein the venting pipe is adapted to vent any gas escaping from the apparatus to the bubbler tank wherein the gas is scrubbed and vented on the exterior of the

5 apparatus step (222). The method (200) also includes removing, by a de-humidifier, moisture generated through the bubbler tank to make the engine dry before the hydrogen gas reached to the engine.

[0064] The method (200) also includes positioning, an upper diverter plate at a top of the reaction chamber, wherein the upper diverter plate includes the plurality of  
10 apertures larger than the plurality of holes. Additionally, the method (200) includes reducing, the stripping of the electrodes and to ensure that iron hydroxide and chromium hydroxide are not produced in the electrolytic reactions by adding hydrogen phosphate. Once a phosphate barrier is created on the cells, the solution may be flushed and replaced with a solution of dilute potassium hydroxide or  
15 sodium hydroxide for example 6 to 10 grams potassium hydroxide and 5 to 8 grams sodium hydroxide per one Liter of pure, demineralized, or distilled water.

[0065] Various embodiments of the present disclosure provides an apparatus for hydrogen generation in an economic way. The apparatus disclosed in the present disclosure facilitates energy harnessing from the engine associated battery and  
20 alternator with no additional carbon footprint. The apparatus achieves a considerable reductions in fuel utilization and detrimental emissions in a vehicle. The hydrogen generation unit disclosed in the present disclosure generate hydrogen gas and delivers of significant quantities of hydrogen gas to internal combustion (IC) engine.

25 [0066] Further, the apparatus disclosed in the present disclosure provides a protective agent such as hydrogen phosphate to counter electrolytic erosion of the electrodes. The DC power supply unit disclosed in the present disclosure helps in sustaining a controlled electrolysis over extended durations which may deliver precise voltage and current quantities to the cells while preventing electrolysis from  
30 spiraling out of control. Also, the de-humidifier disclosed in the present disclosure improves the efficiency and longevity of the hydrogen generation unit and contributes to the overall efficacy of the HHO application in diesel engines.

5 [0067] While specific language has been used to describe the disclosure, any limitations arising on account of the same are not intended. As would be apparent to a person skilled in the art, various working modifications may be made to the method (250) in order to implement the inventive concept as taught herein.

[0068] The figures and the foregoing description give examples of  
10 embodiments. Those skilled in the art will appreciate that one or more of the described elements may well be combined into a single functional element. Alternatively, certain elements may be split into multiple functional elements. Elements from one embodiment may be added to another embodiment. For example, the order of processes described herein may be changed and are not  
15 limited to the manner described herein. Moreover, the actions of any flow diagram need not be implemented in the order shown; nor do all of the acts need to be necessarily performed. Also, those acts that are not dependent on other acts may be performed in parallel with the other acts. The scope of embodiments is by no means limited by these specific examples.

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**WE CLAIM:**

1. An apparatus (100) for hydrogen generation comprising:

a hydrogen generating unit (102) comprising:

an electrolyte water tank (104) comprising:

characterized in that,

an electrolyte supply line (106) and an electrolyte return line (108) positioned at a base of the electrolyte water tank (104), wherein the electrolyte supply line (106) is adapted to transfer electrolyte to the electrolyte water tank (104);

an electrolyser assembly (110) connected to the electrolyte supply line (106) and adapted to pass the electrolyte through a plurality of electrode plates (112) to the electrolyte water tank (104) in a turbulent manner via an exit nipple (114),

wherein the plurality of electrode plates (112) cover a pre-defined area with a pre-defined spacing between two consecutive electrode plates, wherein each electrode plate of the plurality of electrode plates (112) comprises a predefined surface area,

wherein each of the plurality of electrode plate (112) comprises an aperture (114a) at a corner of the electrode plate to reduce the charge concentration at the corners of the electrode plates and a central aperture (114b), wherein the central aperture (114b) is supported by a plurality of symmetrical apertures (114c) to bind the plurality of electrode plates (112) in an array with a plurality of bolts (154), and

wherein the plurality of electrode plates (112) is assembled in a plurality of configurations of a plurality of cells, wherein a predetermined cells of the plurality of cells are adapted to connect with an anode (120a) and a cathode (120b);

a turbulence inducing unit (118) connected with the electrolyser assembly (110), wherein the turbulence inducing (118) instrument is a combination of a high-pressure water flow arrangement and a flow reducing diverter plate (122),

wherein the flow reducing diverter plate (122) comprises a plurality of first predetermined holes (140a) on two sides and a plurality of second predetermined holes (140b) beneath a middle of the flow reducing diverter plate (122);

a reaction chamber (124) comprised of a moulded polypropylene plastic is connected to the flow reducing diverter plate (122), wherein the reaction chamber (124) comprises a plurality of electrolyte pressure chambers positioned beneath the flow reducing diverter plate (122);

a manifold (126) connected to the electrolyte supply line (106), wherein the manifold is (126) adapted to pump water at a predetermined pressure to create turbulence between the plurality of electrode plates (112);

a plastic chamber (128) connected with the manifold (126), wherein the plastic chamber (128) comprises a plurality of electrodes (130) positioned on at least one of the same side or on opposite side of the electrolyser assembly (110),

wherein the plurality of the electrode plates (112) are configured using first double cell (130a), single cell (130b) and double cells (130c) use same side electrodes and the electrode port provided on the opposite side of the plastic chamber (128) is provided for a plurality of double cells configuration;

a power supply unit (132) connected with the electrolyte water tank (104) and configured to use a forced convection system and a predefined capacity water tank,

wherein the power supply unit (132) is a direct current power

supply with an anode point and a cathode point, arranged to pass a current of a pre-defined amperes to the system at a predefined current range, and

wherein the hydrogen generating unit (102) after a predefined period of constant operation, stabilizes at a pre-determined temperature with a predetermined amount of electrolyte consumption for each four hours.

a venting pipe (134) connected with the hydrogen generating unit (102), wherein the venting pipe (134) is positioned atop of the hydrogen generating unit (102) via an integrated electrolyte tank and a bubbler tank (136),

wherein the venting pipe (134) is adapted to vent any gas escaping from the apparatus to the bubbler tank (136) wherein the gas is scrubbed and vented on the exterior of the apparatus.

2. The apparatus (100) as claimed in claim 1, wherein the hydrogen generating unit (102) comprises an electrolyte pump assembly positioned on an external wall of the hydrogen generating unit (102) and is configured to:

receive electrolyte from the electrolyte supply line (106); and

transmit the electrolyte to the manifold for feeding into a base of electrolyser assembly.

3. The apparatus (100) as claimed in claim 1, wherein the electrode plate comprises an area of  $18458 \text{ cm}^2$  with a spacing of 2 mm, for  $577 \text{ cm}^2$  surface area of each electrode plate.

4. The apparatus (100) as claimed in claim 1, wherein the plurality of electrode plates (112) is rectangle in shape and comprises a partial circular cutout (138a) at three corners of the electrode plate (112) and one cutout (138b) at one remaining corner of the plate.

5. The apparatus (100) as claimed in claim 1, wherein the plurality of holes comprises a pipe with an inner diameter of 9 mm at top side of the pipe and one or more pipes of 4 mm

inner diameter at the bottom side of the one or pipes for the flow of pressurized water through the reaction chamber (124).

6. The apparatus (100) as claimed in claim 1, wherein the hydrogen generation unit (102) comprises an upper diverter plate (142) positioned at a top of the reaction chamber (124), wherein the upper diverter plate (142) comprises a plurality of first apertures (116a, 116b) larger than the plurality of holes.

7. The apparatus (100) as claimed in claim 1, wherein the electrolyte supply line (106) comprises a filter adapted to be replaced or cleaned periodically.

8. The apparatus (100) as claimed in claim 1, wherein the manifold (124) is configured to clean and flush the plurality of electrolyte pressure chambers during service to clean a plurality of contaminants, wherein the plurality of contaminants comprises ferrous hydroxide and chromium hydroxide.

9. The apparatus (100) as claimed in claim 1, comprises a de-humidifier connected with the bubbler tank (136) and adapted to remove moisture generated through the bubbler tank (136) to make the engine dry before the hydrogen gas reached to the engine.

10. A method (200) for assembling the apparatus for hydrogen generation comprising:

characterized in that,

transferring, by an electrolyte supply line of a hydrogen generating unit, electrolyte to an electrolyte water tank; (202)

covering, a pre-defined area with a pre-defined spacing between two consecutive electrode plates by the plurality of electrode plates, wherein each electrode plate of the plurality of electrode plates comprises a predefined surface area; (204)

providing, a corner aperture and a central aperture at of each of the electrode plate of plurality of electrode plate, wherein the central aperture is supported by a plurality of symmetrical apertures to bind the plurality of electrode plates in an array with a plurality of bolts; (206)

assembling, the plurality of electrode plates in a plurality of configurations of a

plurality of cells, wherein a predetermined cells of the plurality of cells are adapted to connect with an anode and a cathode; (208)

providing, the turbulence inducing instrument as a combination of a high-pressure water flow arrangement and a flow reducing diverter plate; (210)

providing, a plurality of first predetermined holes on two sides and a plurality of second predetermined holes beneath a middle of the flow reducing diverter plate; (212)

providing, a reaction chamber comprised of a moulded polypropylene plastic is connected to the flow reducing diverter plate, wherein the reaction chamber comprises a plurality of electrolyte pressure chambers positioned beneath the flow reducing diverter plate; (214)

pumping, by a manifold, water at a predetermined pressure to create turbulence between the plurality of electrode plates; (216)

positioning, a plurality of electrodes of a plastic chamber on at least one of the same side or on opposite side of the electrolyser assembly, wherein the plurality of the electrode plates using double, single and double cells use same side electrodes and the electrode port provided on the opposite side of the plastic chamber; (218)

supplying, by a power supply unit, power to use a forced convection system and a predefined capacity water tank, wherein the power supply is a direct current power supply with an anode point and a cathode point, wherein the anode point, and the cathode point is arranged to pass a current of a pre-defined amperes to the system at a predefined current range; (220) and

positioning, a venting pipe atop of the hydrogen generating unit via an integrated electrolyte tank and a bubbler tank, wherein the venting pipe is adapted to vent any gas escaping from the apparatus to the bubbler tank wherein the gas is scrubbed and vented on the exterior of the apparatus. (222)

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**Dated this 26th day of February 2024**

Signature



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5           AN APPARATUS FOR HYDROGEN GENERATION AND A METHOD  
              THEREOF

ABSTRACT

10       An apparatus (100) for hydrogen generation is disclosed. The apparatus includes a  
hydrogen generating unit (102) and a venting pipe (134). The hydrogen generating  
unit includes electrolyte water tank (104) including, an electrolyte supply line (106)  
and an electrolyte return line (108) transfers electrolyte to the electrolyte water tank,  
an electrolyser assembly (110) passes the electrolyte through a plurality of electrode  
plates (112) via an exit nipple (114), a turbulence inducing unit (118) is a  
combination of a high-pressure water flow arrangement and a flow reducing  
15       diverter plate (122), a reaction chamber (124) includes a plurality of electrolyte  
pressure chambers, a manifold (126) pumps water at a predetermined pressure, a  
plastic chamber (128) includes a plurality of electrodes (130), a power supply unit  
(132) uses a forced convection system and a predefined capacity water tank. The  
venting pipe vents any gas escaping from the apparatus to the bubbler tank (136).

20       FIG. 1

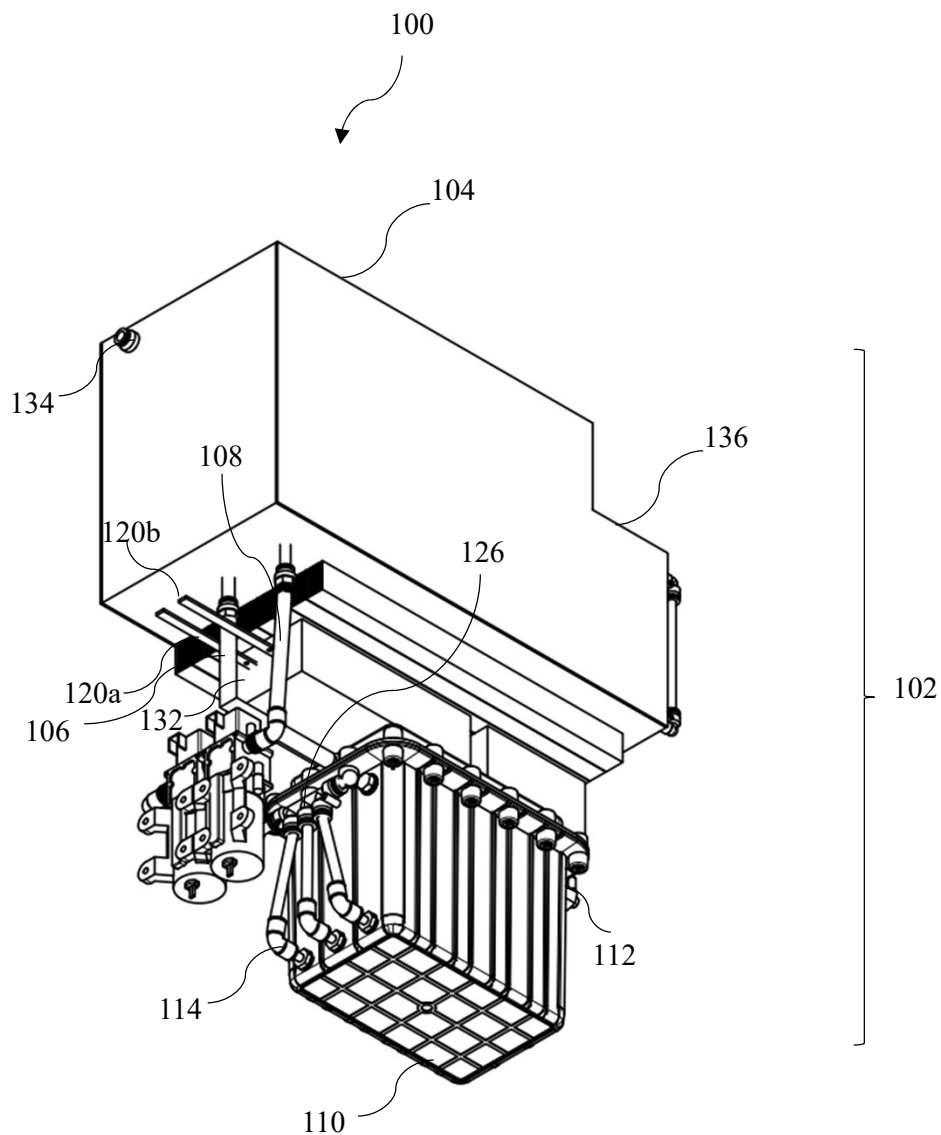


FIG. 1

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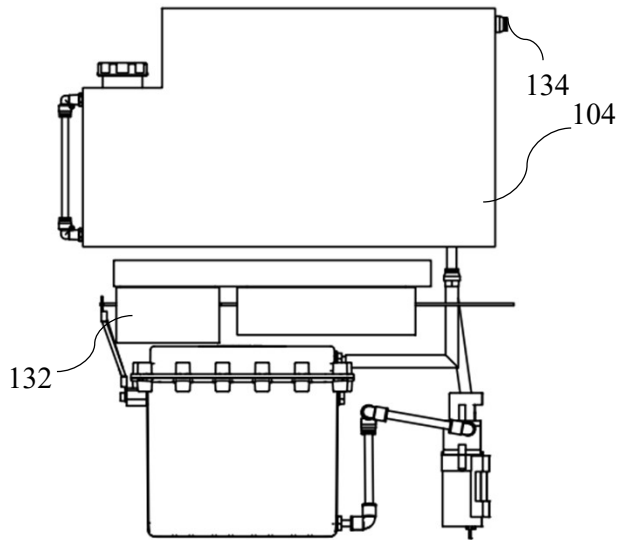


FIG. 2a

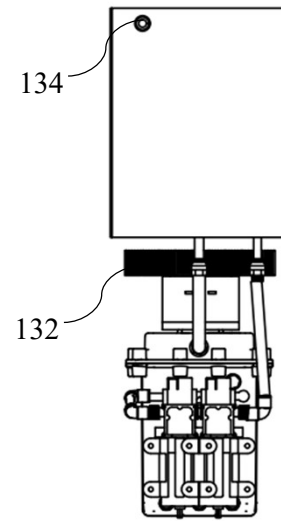


FIG. 2b

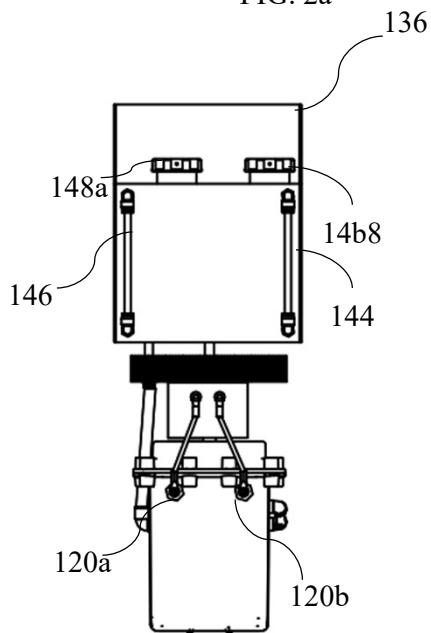


FIG. 2c

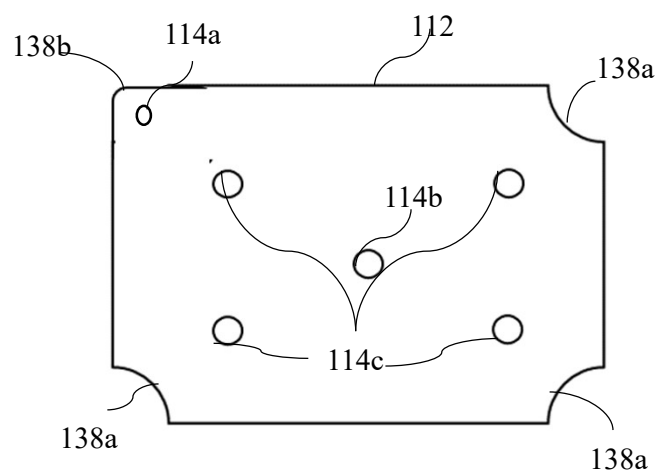


FIG. 2d

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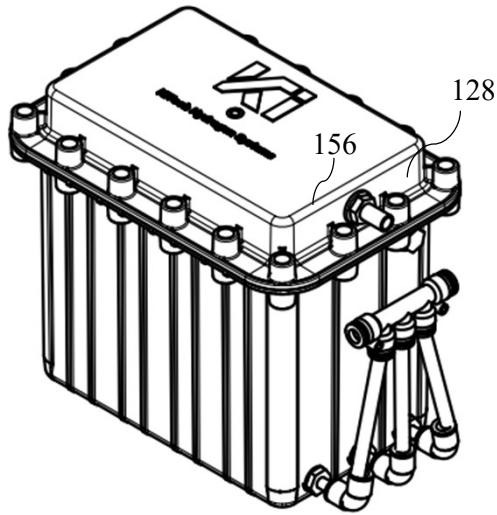


FIG. 3a

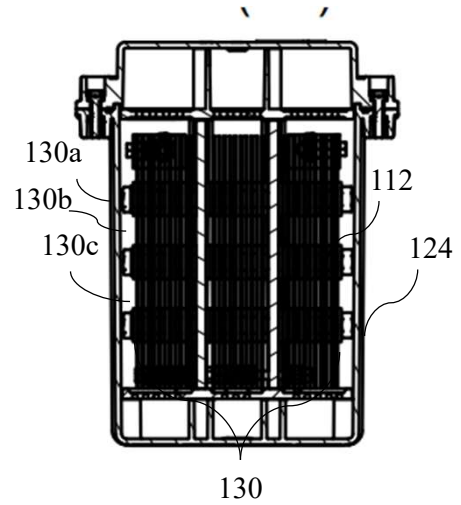


FIG. 3b

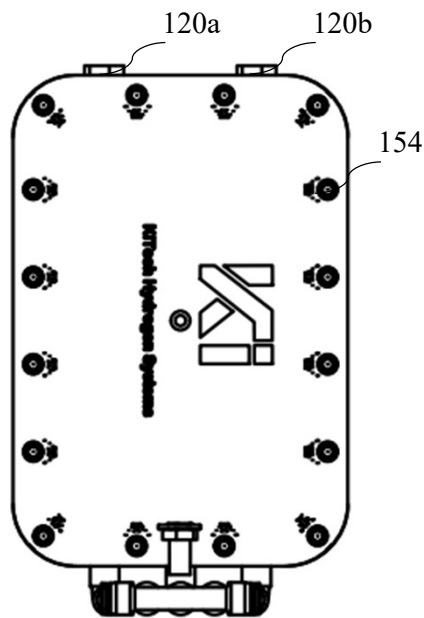


FIG. 3c

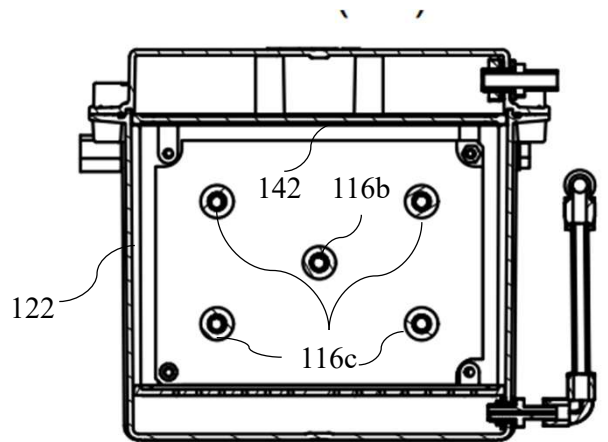
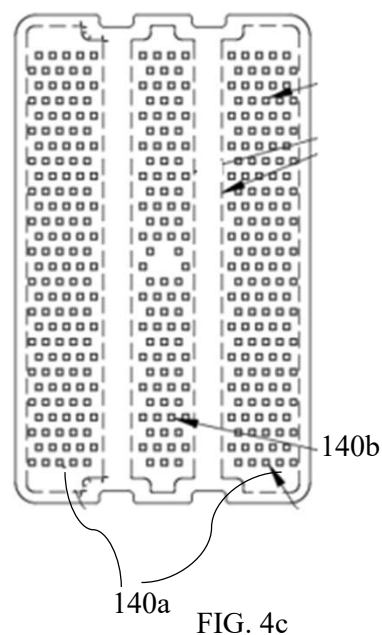
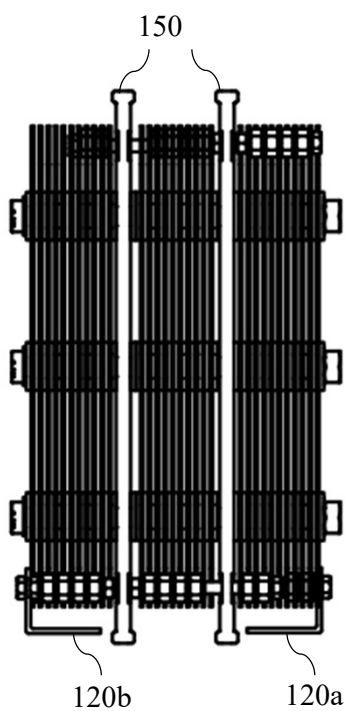
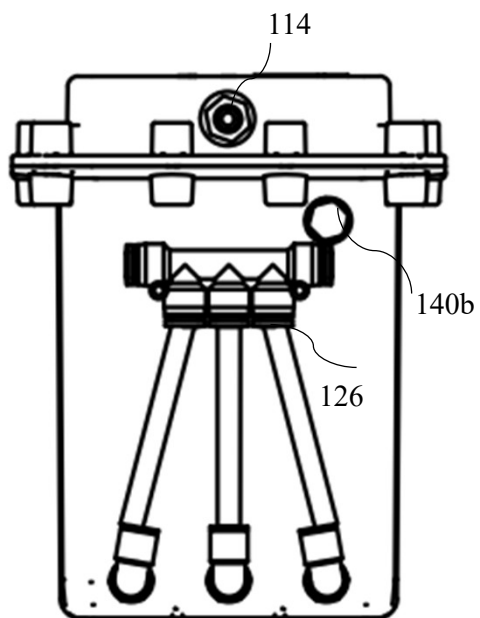


FIG. 3d

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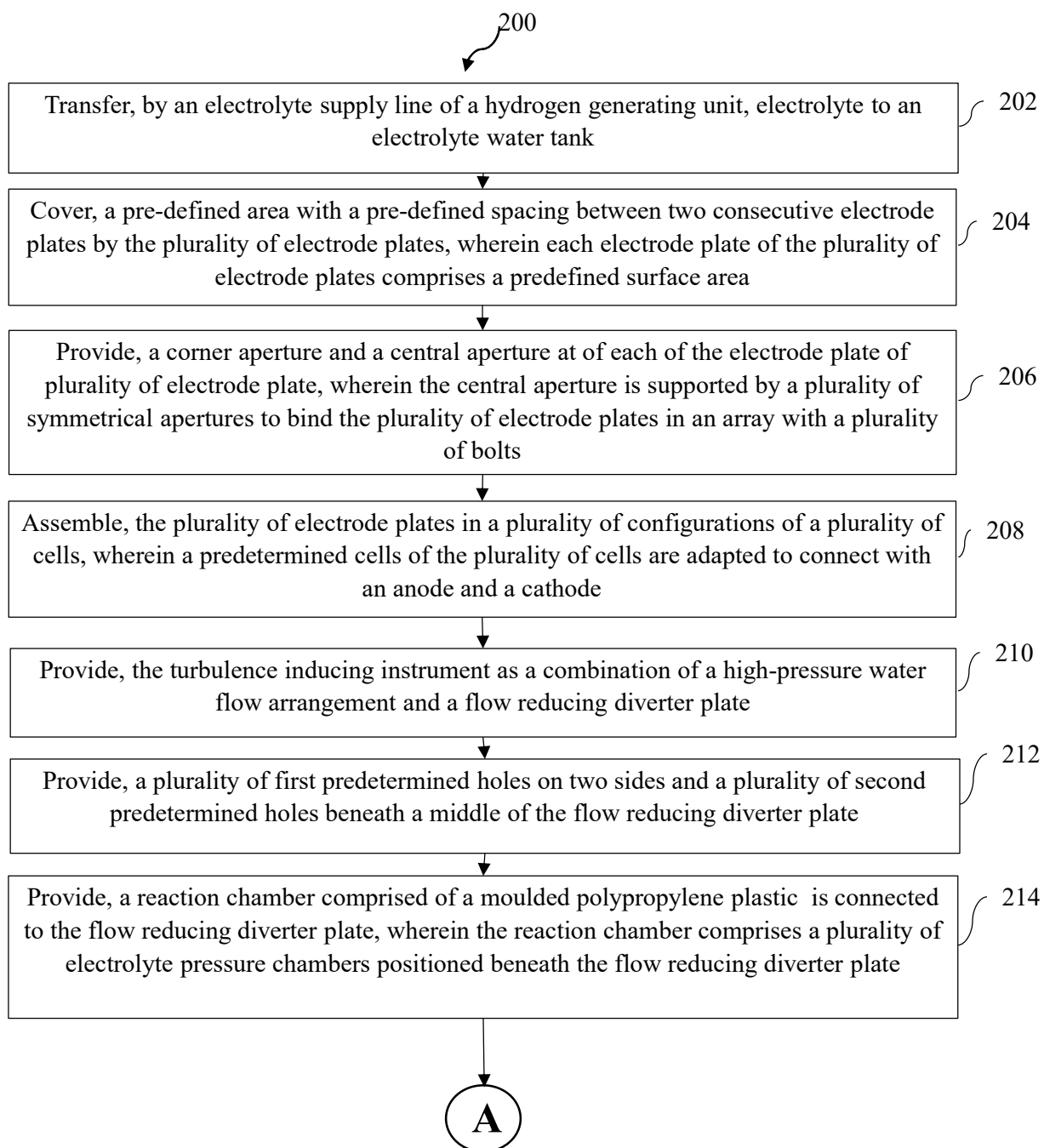


FIG. 5a

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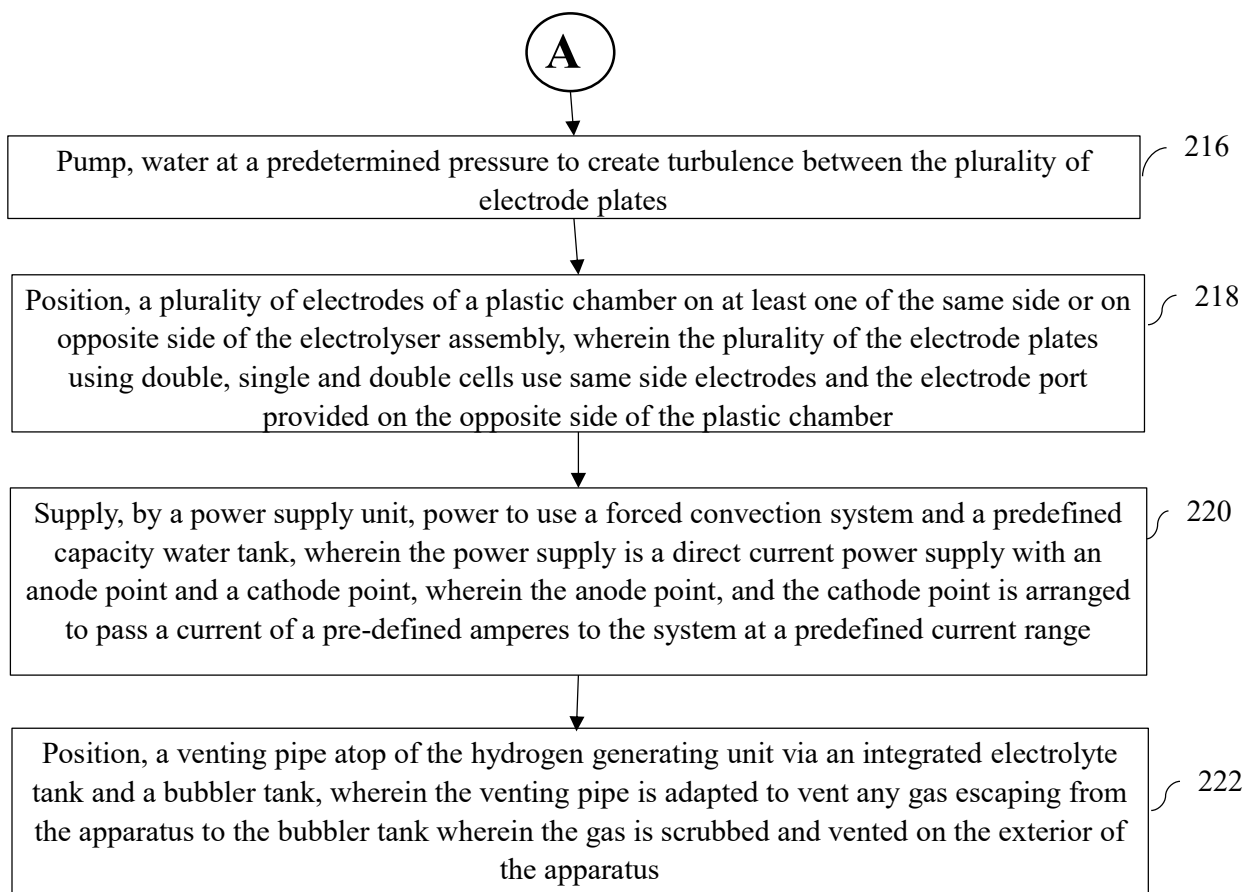


FIG. 5b

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